SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Game Based Learning Assessment

Code of Conduct:

I K.Hemakshitha/ 192425088 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.Hemakshitha

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Answer All the Questions

Q. No.	Scenario / Game Question	Platform	Link
1	**AI Maze Escape:** You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. **Challenge:** Find the shortest path and record the number of steps taken.	Scratch	https://scratch.mit.edu
2	**N-Queens Challenge:** You are an AI chess strategist. **Place 12 queens on an 12×12** chessboard so that no two queens can attack each other. **Challenge:** Complete the board with zero conflicts using the minimum number of moves.	Online N-Queens Game	https://www.n-queens.com
3	**Water Jug Puzzle:** You are an AI agent solving a water jug problem. You have two jugs with capacities of **11 litres and 9 litres**. Neither jug has measurement markings. **Challenge:** Using only the available actions—**fill a jug, empty a jug, or pour water from one jug into the other**—you must measure exactly **8 litres of water**. **Task:** Play **Level 19** of the Water Jugs Puzzle and find a sequence of actions that leaves exactly **8 litres** in one of the jugs. Try to solve the puzzle using the **minimum number of moves**.	Tic-Tac-Toe	https://playtictactoe.org
4	**Connect Four AI Challenge:** You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. **Challenge:** Plan your moves carefully and defeat the AI opponent.	Connect Four	https://www.mathsisfun.com/games/connect4.html

5	**8-Puzzle AI Challenge:** You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. **Challenge:** Solve the puzzle using the minimum possible number of moves.	8-Puzzle Game	https://8puzzle.org/

Structure of the Report – Game-Based Learning

S.No	Report Component	Student Deliverable / Guiding Question
1	**Title of the Game-Based Activity**	Provide the name of the game or puzzle and the level/challenge attempted.
2	**Game Platform / Link**	Mention the platform or website used and provide the corresponding game link.
3	**Objective of the Activity**	Explain the objective of the game and the target that needs to be achieved.
4	**Problem Statement**	Describe the problem or challenge given in the game. What is the student required to solve?
5	**AI Concept / Domain**	Identify the AI concept involved, such as State-Space Search, BFS, DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	**Initial State and Goal State**	Identify and describe the initial state and the goal state of the problem.
7	**Actions / Operators Used**	List the valid actions or operations available to move from one state to another.
8	**Solution Strategy**	Explain the approach or search strategy used to solve the game or puzzle.
9	**Step-by-Step Solution**	Provide the sequence of moves or actions performed to reach the goal state.
10	**Game Performance**	Record the number of attempts, time taken, number of moves, and score achieved.
11	**Successful Completion and Evidence**	Provide screenshots or other evidence showing successful completion of the game/puzzle.
12	**Efficiency and Optimality**	Analyse whether the solution was optimal or efficient. Can the puzzle be solved using fewer moves or less time?
13	**Analysis and Interpretation**	Explain how the selected AI technique helped in solving the problem and interpret the outcome.
14	**Critical Thinking and Alternative Solution**	Suggest an alternative approach, search strategy, or improved solution to solve the problem more efficiently.
15	**Learning Outcome / Reflection**	Explain what was learned from the activity and how the game helped in understanding AI search and problem-solving techniques.
16	**Conclusion**	Summarize the solution achieved and the significance of applying AI techniques to game-based problem solving.
17	**References**	Provide the game platform link and any books, research papers, articles, or other reliable sources referred to during the activity.

TITLE : AI Maze Escape

TITLE OF THE GAME-BASED ACTIVITY: AI Maze Escape

GAME PLATFORM / LINK:

LINK: https://scratch.mit.edu/projects/10128431/editor/

OBJECTIVES :

To help the AI robot move from the START point to the GOAL.

To find the shortest path while avoiding walls and obstacles.

To complete the maze using the minimum number of steps.

PROBLEM STATEMENT :

The AI robot starts from the **START** point and has to reach the **GOAL** by finding the shortest path. The robot cannot move through walls, so it has to choose the correct route. The main challenge is to reach the goal in the fewest steps possible without getting stuck.

AI CONCEPT /DOMAIN :

This activity is based on Pathfinding in Artificial Intelligence. It shows how an AI system can search for the best route from one place to another.

Concepts used:

Pathfinding

Breadth-First Search (BFS)

State Space Search

INITIAL STATE:

The robot is at the START position.

The maze contains open paths and walls.

The goal has not been reached.

GOAL STATE:

- The robot reaches the **GOAL**.
- The shortest path is found.
- The total number of steps is recorded.

ACTIONS / OPERATORS USED :

Move Up

Move Down

Move Left

Move Right

Avoid walls

Continue moving until the GOAL is reached.

SOLUTION STRATEGY:

I started from the START position and checked all the possible paths. Whenever I reached a wall, I chose another direction. I continued moving through the open paths until I reached the GOAL. While solving the maze, I tried to follow the shortest route and counted the total number of steps taken.

STEP BY STEP SOLUTION:

Start from the START position.

Move through the open path.

Check all possible directions.

Avoid walls and blocked paths.

Continue moving toward the goal.

Follow the shortest available route.

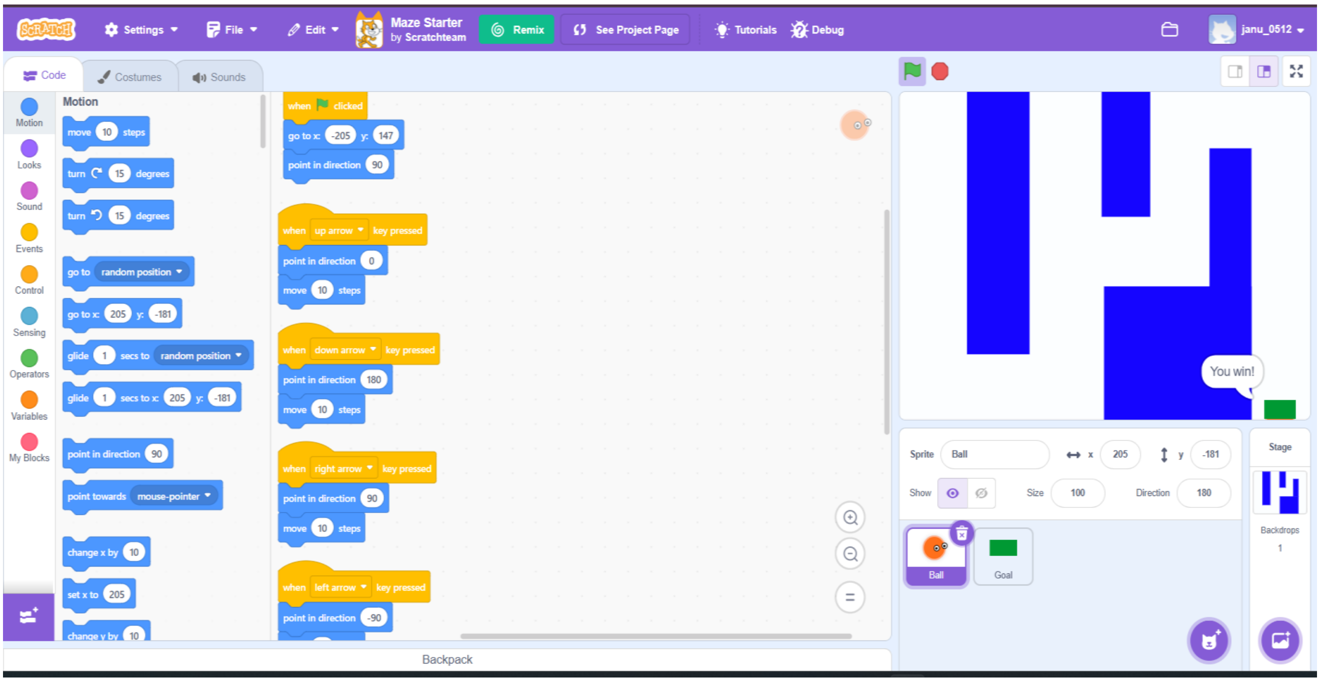
Reach the GOAL.

Record the total number of steps taken.

PERFORMANCE:

PARAMETERS	**RESULT**
Attempts	1
Time taken	2-3 min
No.of steps	10
Goal reached	Yes
Shortest path	Yes

SUCCESSFUL COMPLETION AND EVIDENCE:



EFFICIENCY AND OPTIMALITY:

The maze was solved using the shortest available path, reducing unnecessary movements. The solution was efficient because the robot reached the goal with minimum steps and without revisiting blocked paths.

ANALYSIS AND INTERPRETATION:

This activity demonstrates how AI can solve pathfinding problems by exploring different routes and selecting the best one. It shows the importance of search algorithms in helping an AI agent make decisions and find the shortest path in a maze.

CRITICAL THINNKING AND ALTERNATIVE SOLUTION:

An alternative approach is to use the A* Search Algorithm. Unlike BFS, A* uses a heuristic to estimate the distance to the goal, allowing it to find the shortest path more efficiently in larger or more complex mazes.

LEARNING OUTCOMES:

I learned how AI can find the shortest path in a maze.

I understood the importance of planning before making a move.

This activity improved my logical thinking and problem-solving skills.

I also learned how search algorithms help AI make better decisions.

CONCLUSION:

The AI Maze Escape activity was completed successfully by guiding the robot from the START position to the GOAL using the shortest path. This activity helped me understand the practical application of AI search algorithms in solving pathfinding problems and improving problem-solving skills.

REFERENCES:

1. Scratch – https://scratch.mit.edu
2. Russell, S., & Norvig, P. Artificial Intelligence: A Modern Approach (4th Edition).
3. Breadth-First Search (BFS) – https://en.wikipedia.org/wiki/Breadth-first_search

TITLE : N-Queens Challenge

TITLE OF THE GAME-BASED ACTIVITY: N-Queens Challenge – 12×12 Chessboard

GAME PLATFORM / LINK :

PLATFORM : Online N-Queens Game

LINK : https://www.n-queens.com

OBJECTIVE OF THE ACTIVITY:

The objective is to place 12 queens on a 12×12 chessboard such that no two queens attack each other. This means that no two queens can be in the same row, column, or diagonal. We want to find the minimum number of moves to reach a conflict free arrangement.

PROBLEM STATEMENT:

The N-Queens Challenge is a puzzle where you have to place 12 queens on a 12×12 chessboard. The goal is to make sure that no two queens can attack each other. This means that no two queens should be in the same row, column, or diagonal. The puzzle is solved only when all 12 queens are placed without any conflicts. It requires careful thinking and trying different positions until the correct arrangement is found. This challenge helps improve logical thinking and problem-solving skills and is often used to explain AI concepts like Constraint Satisfaction and Backtracking Search.

AI CONCEPTS / DOMAIN:

The N-Queens Challenge belongs to the field of Artificial Intelligence (AI), specifically under the Constraint Satisfaction Problem (CSP) domain. In this puzzle, the goal is to place 12 queens on a 12×12 chessboard while following a set of rules, or constraints. AI algorithms solve the puzzle by checking different positions, avoiding conflicts, and finding a valid solution. The challenge is commonly solved using techniques such as Backtracking Search, State Space Search, and Heuristic Search. It is widely used to demonstrate how AI can solve complex problems through logical reasoning and systematic decision-making.

INITIAL STATE AND GOAL STATE :

INITIAL STATE:

The 12×12 chessboard is empty.

No queens are placed.

Every position on the board is available.

GOAL STATE:

The 12×12 chessboard is empty.

No queens are placed.

Every position on the board is available.

ACTIONS/ OPERATORS USED :

Place a queen on an empty square.
Check for row conflicts.
Check for column conflicts.
Check for diagonal conflicts.
Remove a queen if a conflict occurs (Backtracking).
Try another position.
Repeat the process until all queens are safely placed.

SOLUTION STRATEGY :

I started by placing one queen in the first row. Before placing the next queen, I checked whether the selected position was safe. If a conflict occurred, I removed the previously placed queen and tried another position. This process was repeated until all 12 queens were placed without any conflicts. The Backtracking Search technique helped in finding a valid solution efficiently.

STEP BY STEP SOLUTION:

QUEEN	**POSITION**
Q1	(1,1)
Q2	(2,3)
Q3	(3,5)
Q4	(4,8)
Q5	(5,10)
Q6	(6,12)
Q7	(7,6)
Q8	(8,11)
Q9	(9,2)
Q10	(10,7)
Q11	(11,9)
Q12	(12,4)

STEPS FOLLOWED:

Start with an empty 12×12 chessboard.

Place the first queen in a safe position.

Move to the next row.

Check row, column, and diagonal conflicts.

Place the queen if the position is safe.

If a conflict occurs, remove the previous queen and try another position.

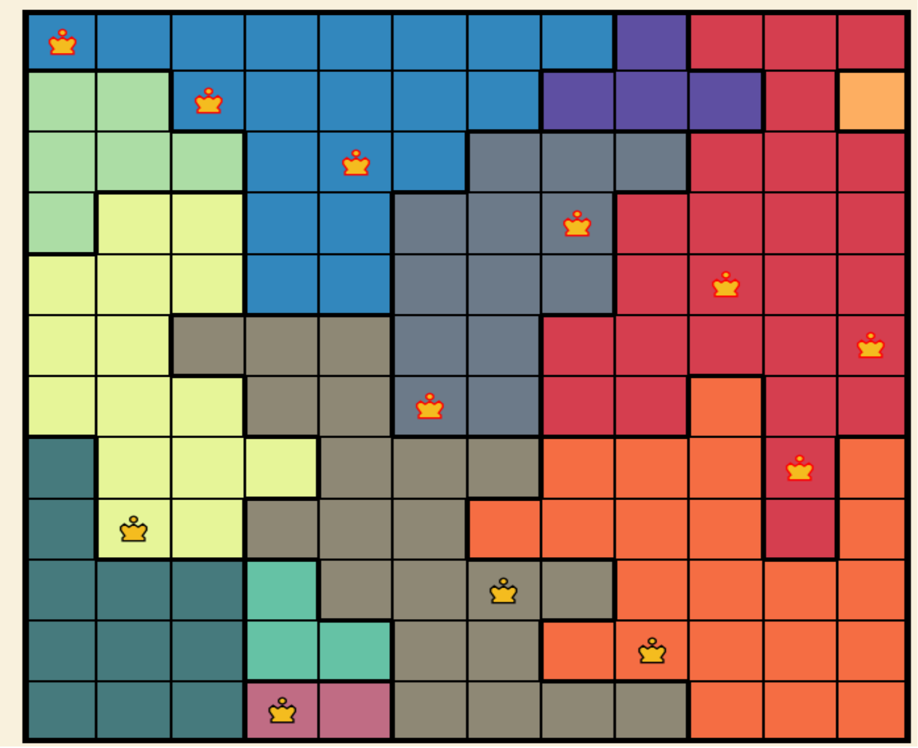
Repeat the process until all 12 queens are placed safely.

Verify that no queens attack each other.

GAME PERFORMANCE:

PARAMETER	RESULT
Attempts	**1**
Time taken	**8 min**
No.of moves	**12**
Conflicts	**0**
Puzzle status	**Sucessfully completed**

SUCCESSFUL COMPLETION AND EVIDENCE :



EFFICIENCY AND OPTIMALITY:

The solution was efficient because every queen was placed without violating any constraints. The Backtracking algorithm reduced unnecessary searches by removing incorrect placements and trying alternative positions. Although many valid solutions exist, the obtained arrangement satisfies all the puzzle constraints.

ANALYSIS AND INTERPRETATION:

This activity demonstrates how Artificial Intelligence solves Constraint Satisfaction Problems. The Backtracking algorithm checks every placement carefully and only continues when the current position is valid. This reduces unnecessary work and ensures that a correct solution is found. The activity also highlights the importance of logical reasoning and systematic problem-solving.

CRITICAL THINKING AND ALTERNATIVE SOLUTION:

Instead of Backtracking, the puzzle can also be solved using:

Min-Conflicts Heuristic

Genetic Algorithm

Hill Climbing

Among these methods, the Min-Conflicts Heuristic is often more efficient for solving larger N-Queens problems because it quickly reduces the number of conflicts.

LEARNING OUTCOME / REFLECTION :

This activity helped me understand how AI solves complex problems by applying constraints and searching for valid solutions. I learned about Constraint Satisfaction Problems (CSP) and the Backtracking Search algorithm. It also improved my logical thinking, analytical skills, and decision-making abilities.

CONCLUSION :

The 12×12 N-Queens Challenge was completed successfully by placing all 12 queens without any conflicts. The activity demonstrated the practical use of AI techniques such as Constraint Satisfaction and Backtracking Search. It also enhanced my understanding of logical reasoning and efficient problem-solving methods used in Artificial Intelligence.

REFERENCES:

Online N-Queens Game: https://www.n-queens.com
Russell, S., & Norvig, P. **Artificial Intelligence: A Modern Approach (4th Edition).**
GeeksforGeeks – N-Queen Problem: https://www.geeksforgeeks.org/n-queen-problem/

TITLE : WATER JUG PROBLEM

TITLE OF THE GAME-BASED ACTIVITY: Water Jug Puzzle – Level 19 (11L and 9L Jugs)
GAME PLATFORM / LINK:

LINK: https://playtictactoe.org

OBJECTIVE OF THE ACTIVITY:
The objective of this activity is to measure exactly 8 litres of water using two jugs with capacities of 11 litres and 9 litres. Only three actions are allowed: filling a jug, emptying a jug, and pouring water from one jug to the other. The goal is to achieve the target using the minimum number of moves.

PROBLEM STATEMENT :

In this puzzle, two water jugs with capacities of 11 litres and 9 litres are provided, but neither jug has measurement markings. The challenge is to measure exactly 8 litres of water using only the allowed actions. The solution requires careful planning and selecting the correct sequence of moves to reach the goal.

AI CONCEPT / DOMAIN:

The Water Jug Puzzle is a State Space Search problem in Artificial Intelligence.

AI Concepts Used:

- State Space Search
- Breadth-First Search (BFS)
- Constraint Satisfaction
- Problem Solving

INITIAL STATE:

· Both jugs are empty.
· 11L Jug = 0 litres
· 9L Jug = 0 litres

GOAL STATE:

Measure exactly 8 litres in either the 11L jug or the 9L jug.

ACTIONS/OPERATORS USED:

Fill the 11L jug.
Fill the 9L jug.
Empty the 11L jug.
Empty the 9L jug.
Pour water from the 11L jug into the 9L jug.
Pour water from the 9L jug into the 11L jug.

SOLUTION STRATEGY:

I started with both jugs empty and used the allowed actions to transfer water between the jugs. After every move, I checked the amount of water in each jug. By repeating the process of filling, emptying, and pouring, I eventually measured exactly 8 litres. This method follows the State Space Search approach, where each move creates a new state until the goal is reached.

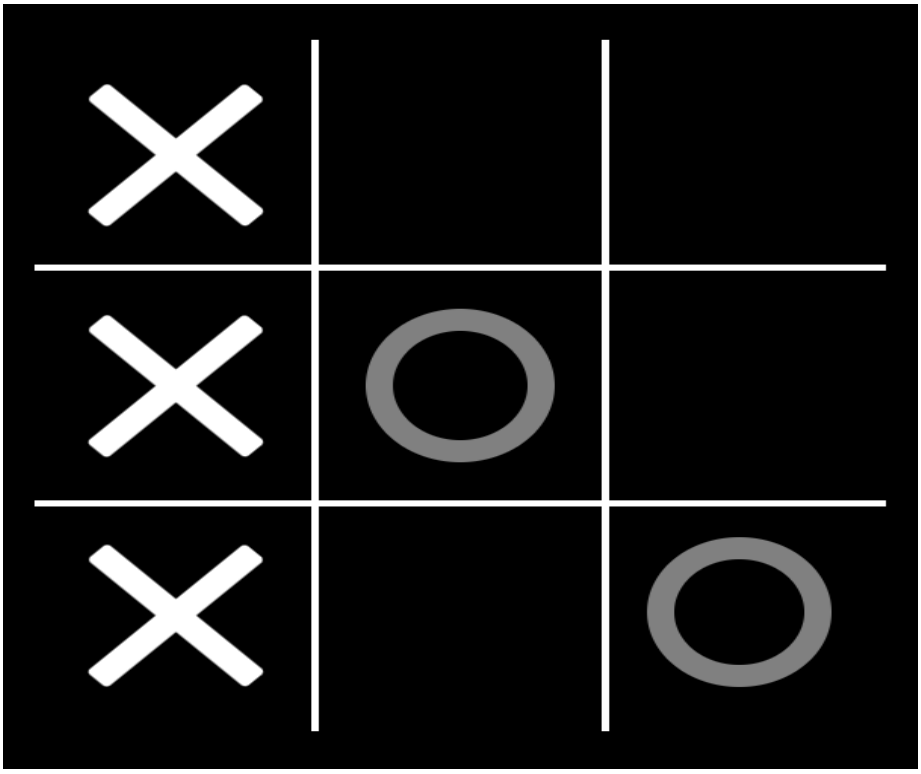
STEP BY STEP SOLUTION :

MOVE	ACTION	STATE(11L,9L)
1	Fill the 11L jug	(8, 9)
2	Pour 11L ->9L	(2, 9)
3	Empty the 9L jug	(2,0)
4	Pour remaining 2L into the 9L jug	(0, 2)
5	Fill the 11L jug	(11, 2)
6	Pour 11L → 9L until full	(4, 9)
7	Empty the 9L jug	(4, 0)
8	Pour 4L into the 9L jug	(0, 4)
9	Fill the 11L jug	(11, 4)
10	Pour 11L → 9L until full	(6, 9)
11	Empty the 9L jug	(6, 0)
12	Pour 6L into the 9L jug	(0, 6)
13	Fill the 11L jug	(11, 6)
14	Pour 11L → 9L until full	(8, 9)

GAME PERFORMANCE :

PARAMETER	**RESULT**
Atempt	1
Time taken	5 min
No.of moves	14
Target Achieved	Yes
Final Result	8 liters measured

SUCCESSFUL COMPLETION AND EVIDENCE :



EFFICIENCY AND OPTIMALITY:

The solution was efficient because it used only the allowed operations and reached the target amount successfully. Each move changed the state of the jugs, and unnecessary actions were avoided. The puzzle can be solved using search algorithms such as Breadth-First Search (BFS) to guarantee the shortest sequence of moves.

ANALYSIS AND INTERPRETATION:

This activity demonstrates how AI solves problems by exploring different states until the goal is reached. Each action changes the amount of water in the jugs, creating a new state. AI algorithms can systematically search these states to find a valid and efficient solution.

CRITICAL THINKING AND ALTERNATIVE SOLUTION:

Another approach to solving the Water Jug Puzzle is using the A* Search Algorithm or Depth-First Search (DFS). These methods also explore possible states but differ in the way they search for the solution. BFS is generally preferred because it finds the shortest sequence of moves.

LEARNING OUTCOME / REFLECTION :

This activity helped me understand how AI solves real-world problems using search techniques. I learned about State Space Search, Breadth-First Search (BFS), and how different actions lead to different states. It also improved my logical thinking and problem-solving skills.

CONCLUSION :

The Water Jug Puzzle was successfully solved by measuring exactly 8 litres using the available actions. The activity demonstrated the practical use of State Space Search and Breadth-First Search (BFS) in Artificial Intelligence. It also improved my understanding of logical reasoning, planning, and decision-making while solving constraint-based problems.

REFERNCCES:

· Water Jug Puzzle: https://playtictactoe.org
· Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
· GeeksforGeeks – Water Jug Problem: https://www.geeksforgeeks.org/water-jug-problem-using-bfs/

TITLE :CONNECT FOUR

TITLE OF THE GAME-BASED ACTIVITY: Connect Four AI Challenge

GAME PLATFORM / LINK:

LINK:https://www.math sisfun.com/games/connect4.html

OBJECTIVE OF THE ACTIVITY:

The objective of this activity is to play Connect Four against an AI opponent and connect four discs of the same color in a row before the AI does. The four discs can be arranged horizontally, vertically, or diagonally. The goal is to use a good strategy to win or prevent the AI from winning.

PROBLEM STATEMENT :

In Connect Four, two players take turns dropping colored discs into a vertical grid. The discs fall to the lowest empty position in the selected column. The challenge is to create a line of four discs while blocking the opponent from forming their own line. Since the opponent is an AI, each move must be planned carefully to improve the chances of winning.

AI CONCEPT / DOMAIN:

Connect Four is based on Game Playing in Artificial Intelligence.

AI Concepts Used:

- Game Playing
- Minimax Algorithm
- Alpha-Beta Pruning
- Decision Making
- Adversarial Search

INITIAL STATE:

· The Connect Four board is empty.
· No discs have been placed.
· The game is ready to begin.

GOAL STATE:

· Connect four discs of the same color in a horizontal, vertical, or diagonal line before the AI.
· Win the game or achieve a draw if neither player can win.

ACTIONS/OPERATORS USED:

· Select a column.
· Drop a disc into the selected column.
· Observe the AI's move.
· Block the AI's possible winning moves.
· Continue playing until someone wins or the game ends in a draw.

SOLUTION STRATEGY:

I started by placing my disc in the center columns because they provide more opportunities to create multiple winning combinations. During the game, I observed the AI's moves and blocked any possible four-disc connections. At the same time, I planned my own moves to create horizontal, vertical, or diagonal lines of four discs.

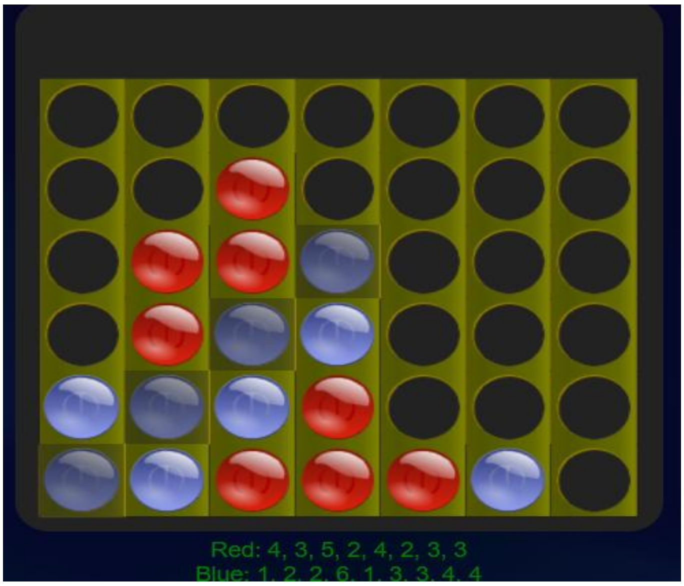
STEP BY STEP SOLUTION :

- Start with an empty board.
- Drop the first disc into the center column.
- Observe the AI's response.
- Continue placing discs while creating possible winning lines.
- Block the AI whenever it gets close to connecting four discs.
- Keep checking horizontal, vertical, and diagonal patterns.
- Connect four discs to win the game.

GAME PERFORMANCE :

PARAMETER	RESULT
Attempt	1
Time taken	5 min
Outcome	Win
AI Difficulty	Medium

SUCCESSFUL COMPLETION AND EVIDENCE :



EFFICIENCY AND OPTIMALITY:

The strategy was effective because the center columns were used to increase the chances of creating multiple winning combinations. Blocking the AI's moves at the right time helped avoid losing. This approach demonstrates efficient decision-making similar to AI game-playing algorithms.

ANALYSIS AND INTERPRETATION:

This activity demonstrates how AI uses decision-making techniques to play strategic games. The AI predicts possible future moves and chooses the best action using algorithms such as Minimax and Alpha-Beta Pruning. The game also highlights the importance of planning several moves ahead.

CRITICAL THINKING AND ALTERNATIVE SOLUTION:

An alternative strategy is to focus on creating multiple winning opportunities instead of only blocking the opponent. This forces the AI to defend several positions at once, increasing the chance of winning. More advanced AI systems also use Monte Carlo Tree Search (MCTS) for stronger gameplay.

LEARNING OUTCOME / REFLECTION :

Through this activity, I learned how AI makes decisions in competitive games. I understood the importance of planning, predicting the opponent's moves, and selecting the best possible action. The game also improved my strategic thinking and problem-solving skills.

CONCLUSION :

The Connect Four AI Challenge helped me understand the application of Artificial Intelligence in game playing. The activity demonstrated how AI algorithms such as Mini max and Alpha-Beta Pruning are used to make intelligent decisions. It also improved my logical thinking, planning, and decision-making abilities.

REFERNCCES:

- Connect Four Game:https://www.math sisfun.com/games/connect4.html
- Russell, S., & Norvig, P. Artificial Intelligence: A Modern Approach (4th Edition).
- GeeksforGeeks – Mini max Algorithm: https://www.geeksforgeeks.org/minimax-algorithm-in-game-theory/

TITLE : 8- PUZZLE AI CHALLENGE

TITLE OF THE GAME-BASED ACTIVITY: 8-Puzzle AI Challenge
GAME PLATFORM / LINK:

LINK: **https://8puzzle.org/**

OBJECTIVE OF THE ACTIVITY:

The objective of this activity is to arrange the numbered tiles in the correct order by moving one tile at a time into the empty space. The goal is to solve the puzzle using the minimum possible number of moves.

PROBLEM STATEMENT :

The 8-Puzzle consists of eight numbered tiles and one empty space arranged on a 3 × 3 grid. The tiles are initially shuffled, and only one tile adjacent to the empty space can be moved at a time. The challenge is to rearrange the tiles into the correct order while making the fewest possible moves.

AI CONCEPT / DOMAIN:

The 8-Puzzle is a Search Problem in Artificial Intelligence.

AI Concepts Used:

- State Space Search
- A* Search Algorithm
- Heuristic Search
- Best-First Search
- Problem Solving

INITIAL STATE:

· The tiles are placed in a random order.
· One space is empty.
· The puzzle is not solved.

GOAL STATE:
The tiles are arranged in the following order:

1 2 3

4 5 6

7 8

ACTIONS/OPERATORS USED:

· Move the tile Up into the empty space.
· Move the tile Down into the empty space.
· Move the tile Left into the empty space.
· Move the tile Right into the empty space.
· Repeat the moves until all tiles are in the correct order.

SOLUTION STRATEGY:

I started by observing the initial arrangement of the tiles and identified the correct position for each number. I moved one tile at a time into the empty space while trying to reduce the distance between the current arrangement and the goal arrangement. I avoided unnecessary moves and focused on reaching the solution using the minimum number of steps. This approach is similar to the A* Search Algorithm, which selects the most promising move based on a heuristic.

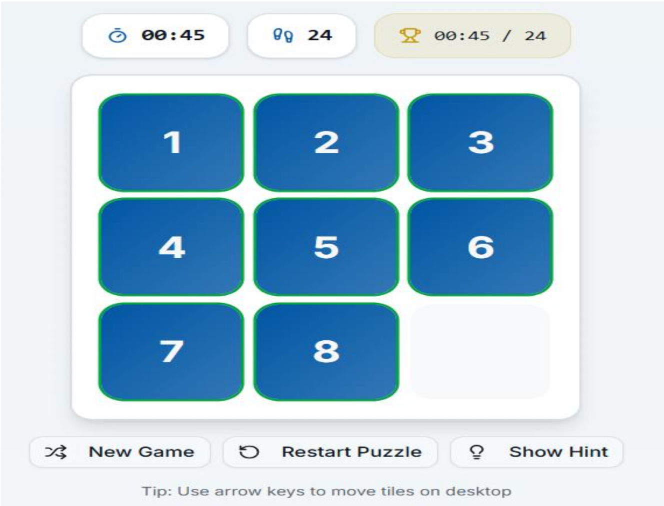
STEP BY STEP SOLUTION :

· Start with the scrambled puzzle.
· Identify the position of the empty space.
· Move an adjacent tile into the empty space.
· Continue placing each tile closer to its correct position.
· Avoid unnecessary backtracking.
· Repeat the process until all tiles are arranged correctly.
· Reach the goal state.

GAME PERFORMANCE :

PARAMETER	RESULT
Attempts	1
Time taken	0.45 sec
Total moves	24
Puzzle solved	yes
Goal reached	yes

SUCCESSFUL COMPLETION AND EVIDENCE :



EFFICIENCY AND OPTIMALITY:

The puzzle was solved by making only valid moves and avoiding unnecessary movements. Using an AI search technique such as A* helps reduce the total number of moves and find an optimal solution efficiently.

ANALYSIS AND INTERPRETATION:

The 8-Puzzle demonstrates how AI solves complex search problems by exploring different states. Each tile movement creates a new state, and the AI evaluates these states to determine the best path to the goal. The A* algorithm is commonly used because it combines the cost of the moves already made with a heuristic estimate of the remaining distance to the goal.

CRITICAL THINKING AND ALTERNATIVE SOLUTION:

Another method for solving the 8-Puzzle is Breadth-First Search (BFS), which guarantees the shortest solution but may require more memory. Other approaches include Depth-First Search (DFS) and Greedy Best-First Search. Among these, A* is generally preferred because it is both efficient and optimal for this type of puzzle.

LEARNING OUTCOME / REFLECTION :

This activity helped me understand how Artificial Intelligence uses search algorithms to solve problems efficiently. I learned about state space search, heuristic functions, and how the A* algorithm finds the shortest solution. The activity also improved my logical thinking, planning, and problem-solving skills.

CONCLUSION :

The 8-Puzzle AI Challenge was completed successfully by arranging the tiles in the correct order. This activity demonstrated the application of A* Search and heuristic techniques in Artificial Intelligence. It also enhanced my understanding of search algorithms and their role in solving real-world optimization problems.

REFERNCCES:

· 8-Puzzle Game:
· Russell, S., & Norvig, P. *Artificial Intelligence: A Modern Approach* (4th Edition).
· GeeksforGeeks – 8 Puzzle Problem: https://www.geeksforgeeks.org/8-puzzle-problem-using-branch-and-bound/

Evaluation Rubrics

Criteria	Percentage (%)	Excellent	Good	Satisfactory	eds Improvem
Understanding of the Problem / Game Objective	10%	Clearly understands the problem and game objective	Understands the objective with minor errors	Partially understands the problem	Does not understand the objective
Application of AI Search / Problem-Solving Technique	15%	Correctly applies the appropriate AI search or problem-solving technique	Applies the technique with minor errors	Shows limited application of the technique	Unable to apply the appropriate technique
Successful Completion of the Game / Puzzle	20%	Successfully solves/completes the game independently	Completes the game with minimal assistance	Partially completes the challenge	Unable to complete the challenge
Efficiency of Solution / Optimal Moves	10%	Solves using optimal or highly efficient moves	Uses a reasonably efficient sequence of moves	Uses several unnecessary moves	Solution is highly inefficient or incomplete
Documentation / Evidence of Completion	15%	Provides clear steps, score, screenshots, and evidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentation or evidence
Analysis and Interpretation of the Solution	15%	Clearly analyses the solution and explains the AI technique and outcome	Provides a good analysis with minor gaps	Provides limited analysis or explanation	Unable to analyse or interpret the solution
Originality and Critical Thinking	15%	Demonstrates excellent independent thinking and explores an effective or alternative solution	Demonstrates good independent thinking	Shows limited critical thinking	Shows little or no independent thinking